



Lab 2

Course Name: Database

Course Code: CSE 301

Student's Full Name:

Student ID:

Question 1. Draw ER Diagram and identify entities, properties, and relationships, then from ER Diagram in change to Database Schema

An industry company requires about employee managements by the computer. The company has many employees. Each employee is recorded with Full Name: Middle Name and First Name, has a unique code, has date of birth, address, gender (gender), will be in charge directly by a manager (also an employee) and belong to a single department. Salaries for employees are determined based on their working positions: full-time employees receive a salary according to company policies, while part-time employees are paid based on their working hours, according to rules pay rates. In the company, there are many departments to manage employee. Each department has a unique code, with a room name, that oversees a certain employee, and records the whole day that the manager starts in charge of the department. Beside, each department can have one or more locations. The company will implement different projects. Therefore, each project records the code and name of the project, the location of the project implementation. The department will be assigned and perform the different projects. Each employee can participate in many different projects, and each project can be participated in by many employees. When an employee joins a project, that employee code, that scheme code, as well as the employee's working time for the project in a week will be recorded. In order to pay more attention to the employees, the company will record some information about the relatives of the employees. The names, dates of birth and relationships of relatives will be recorded.

Question 2. Using MySQL Workbench to create Database, Table, Statements related Database.

1. Give the Database Schema below:

EMPLOYEES (employeeID, lastName, middleName, firstName, dateOfBirth, gender, salary, address, managerID, departmentID)

DEPARTMENT (departmentID, departmentName, managerID, dateOfEmployment)

DEPARTMENTADDRESS (departmentID, address)

PROJECTS (projectID, projectName, projectAddress, departmentID)

ASSIGNMENT (employeeID, projectID, workingHour)

RELATIVE (employeeID, relativeName, gender, dateOfBirth, relationship)

2. Explaining datatype of attributes in the table:

EMPLOYEES (employeeID, lastName, middleName, firstName, dateOfBirth, gender, salary, managerID, departmentID)

#	Attribute	Datatype	Constraints
1	employeeID	VARCHAR(3)	PRIMARY KEY
2	lastName	VARCHAR(20)	NOT NULL
3	middleName	VARCHAR(20)	NULL
4	firstName	VARCHAR(20)	NOT NULL
5	dateOfBirth	DATE	NOT NULL
6	gender	VARCHAR(5)	NOT NULL
7	salary	DECIMAL	NOT NULL
8	address	VARCHAR(100)	NULL
9	managerID	VARCHAR(3)	FORIEGN KEY, NULL
10	departmentID	INT	FORIEGN KEY, NOT NULL

DEPARTMENT (departmentID, departmentName, managerID, dateOfEmployment)

#	Attribute	Datatype	Constraints
1	departmentID	INT	PRIMARY KEY
2	departmentName	VARCHAR(10)	NOT NULL
3	managerID	VARCHAR(3)	FORIEGN KEY
4	dateOfEmployment	DATE	NOT NULL

DEPARTMENTADDRESS (departmentID, address)

#	Attribute	Datatype	Constraints
1	departmentID	INT	PRIMARY KEY, FORIEGN KEY
2	address	VARCHAR(30)	PRIMARY KEY

PROJECTS (projectID, projectName, projectAddress, departmentID)

#	Attribute	Datatype	Constraints
1	projectID	INT	PRIMARY KEY
2	projectName	VARCHAR(30)	NOT NULL

3	projectAddress	VARCHAR(100)	NOT NULL
4	departmentID	INT	FORIEGN KEY

ASSIGNMENT (employeeID, projectID, workingHour)

#	Attribute	Datatype	Contraints
1	employeeID	VARCHAR(3)	PRIMARY KEY, FORIEGN KEY
2	projectID	INT	PRIMARY KEY, FORIEGN KEY
3	workingHour	FLOAT	NOT NULL

RELATIVE (employeeID, relativeName, gender, dateOfBirth, relationship)

#	Attribute	Datatype	Contraints
1	employeeID	VARCHAR(3)	PRIMARY KEY, FORIEGN KEY
2	relativeName	VARCHAR(50)	PRIMARY KEY
3	gender	VARCHAR(5)	NOT NULL
4	dateOfBirth	DATE	NULL
5	relationship	VARCHAR(30)	NOT NULL

3. Showing of Database

EMPLOYEES

lastName	middleName	firstName	employeeID	dateOfBirth	address	gender	salary	managerID	department ID
Dinh	Ba	Tien	123	09/01/1955	731 Tran Hung Dao Q1 TPHCM	Nam	30000	333	5
Nguyen	Thanh	Tung	333	08/12/1945	638 Nguyen Van Cu Q5 TPHCM	Nam	40000	888	5
Tran	Thanh	Tam	453	31/07/1962	543 Mai Thi Luu Ba Dinh Ha Noi	Nam	25000	333	5
Nguyen	Manh	Hung	666	15/09/1952	975 Le Lai P3 Vung Tau	Nam	38000	333	5
Vuong	Ngoc	Quyen	888	10/10/1927	450 Trung Vuong My Tho TG	Nu	55000		1

Le	Thi	Nhan	987	20/06/1931	291 Ho Van Hue Q.PN TPHCM	Nu	43000	888	4
Tran	Hong	Quang	777	29/03/1959	980 Le Hong Phong Vung Tau	Nam	25000	987	4
Bui	Thuy	Vu	999	19/07/1958	332 Nguyen Thai Hoc Quy Nhon	Nam	25000	987	4

DEPARTMENT

departmentID	departmentName	managerID	dateOfEmployment
1	Quan ly	888	19/06/1971
4	Dieu hanh	777	01/01/1985
5	Nghien cuu	333	22/05/1978

DEPARTMENTADDRESS

departmentID	address
1	TP HCM
4	HA NOI
5	NHA TRANG
5	TP HCM
5	VUNG TAU

ASSIGNMENT

employeeID	projectID	workingHour
123	1	22.5
123	2	7.5
123	3	10
333	10	10
333	20	10
453	1	20
453	2	20
666	3	40
888	20	0
987	20	15

PROJECTS

projectID	projectName	projectAddress	departmentID
1	San pham X	VUNG TAU	5
2	San pham Y	NHA TRANG	5
3	San pham Z	TP HCM	5
10	Tin hoc hoa	HA NOI	4
20	Cap Quang	TP HCM	1
30	Dao tao	HA NOI	4

RELATIVE

employeeID	relativeName	gender	dateOfBirth	relationship
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123	Chau	Nu	31/12/1978	Con gai
123	Duy	Nam	01/01/1978	Con trai
123	Phuong	Nu	05/05/1957	Vo chong
333	Duong	Nu	03/05/1948	Vo chong
333	Tung	Nam	25/10/1973	Con trai
333	Quang	Nu	05/04/1976	Con gai
987	Dang	Nam	29/02/1932	Vo chong

Questions:

Using MySQL to

1. Create a new MySQL Database call “managementproject”.
2. Create all of tables in Database with attribute and conditions about data type and constraints (primary key, null, not null) in descriptive above.
3. Insert new records of attributes in all of table.
4. Create constraints (foreign key) in database according to descriptive above.
5. Add a column named “startingDate” with type “DATE” in the “PROJECTS” table. Then change the data type of this column to type year. Next, delete the recently creating column.
6. Change the data type of the column named "gender" in the "EMPLOYEES" and “RELATIVE” table to VARCHAR(3). then, updating CHECK constraints: “Nam”, “Nu” in in the "EMPLOYEES" and “RELATIVE” table.
7. Updating CHECK constraints for attribute “dateOfBirth” in the "EMPLOYEES" and “RELATIVE” table with condition: dateOfBirth < getdate (currently date).
8. Updating UNIQUE constraints in the “DEPARTMENT”.
9. Create a full back up “managementproject” database and drop this database.
- 10.Import “managementproject” database to MySQL with back up before.